EmoViz: Technical Report

Visualising Emotion — the complete analysis pipeline

Technical documentation of the software, experiments and findings

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This document records the technical work; the assessed dissertation is written separately by the student.

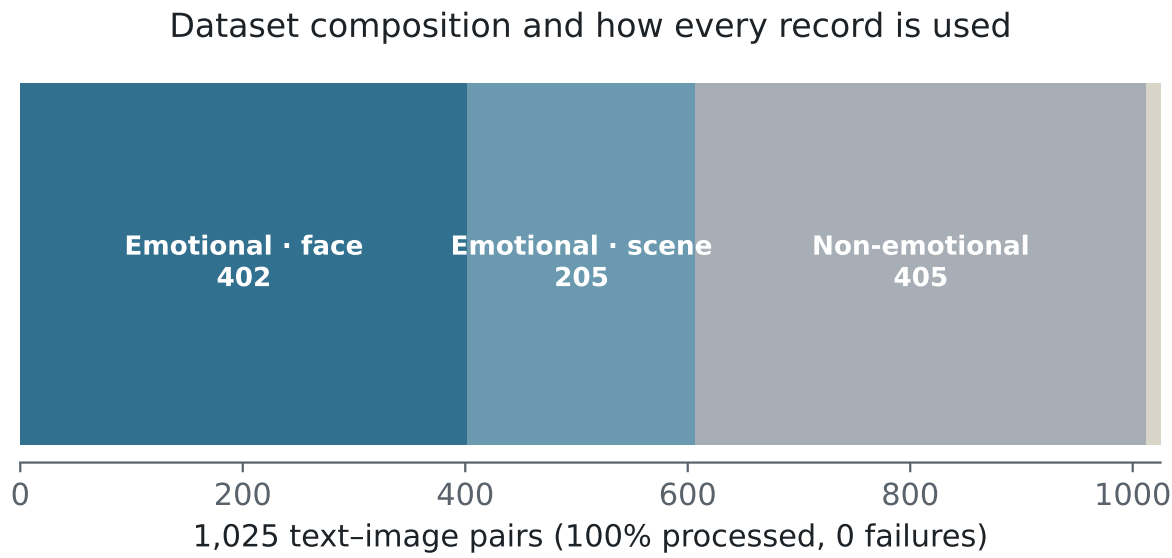
Executive summary

This project asks a question almost untouched in the text-to-image literature: **when an AI turns words about a feeling into a picture, does the picture actually convey that feeling?** Working on a unique dataset of **1,025 public self-reports** — each rendered as a black-and-white line drawing by DALL·E at a public engagement event — we built, validated and verified a complete measurement pipeline. Four findings emerged:

1. **Faces carry emotion; scenes barely do.** Drawings with a readable face agree with the prompt's emotional direction **74%** of the time (family agreement 62%); scene-only drawings manage **43%** (family 36%). Both exceed chance; the gap is large and survives every robustness check.
2. **The pipeline passes its validity test.** For prompts that contain no feeling (40% of the data), measured alignment drops to exactly chance ($z \approx 0$) — the method finds emotion only where emotion exists.
3. **Generative emotion is unstable.** Across 87 prompts drawn 2–42 times each, the model conveys the *same* emotion only 69% of the time; only 32% of repeated prompts are fully consistent across their drawings.
4. **Method matters — and the supervisor's architecture wins.** The describe-then-extract design roughly doubles both emotion-family agreement (49% vs 25%) and top-3 overlap (49% vs 23%) compared with asking CLIP for the emotion directly.

All results are measured against random-pairing null baselines (z-scores 4–16), independently re-derived by an adversarial statistical review, checked on deduplicated prompts, and covered by an automated **28/28-passing test suite**.

1 The dataset — and how every record is used



1,025 text–image pairs from Festival of the Mind 2024 (Sheffield): visitors typed how they felt; DALL·E drew each entry as monochrome line art. Median prompt length is 4 words; mixed feelings ("tired but happy") are common. Key data-engineering facts established:

- **Three incompatible filename schemes** and two lowercase extensions were unified by a robust loader (`emoviz/data.py`) — a naïve loader resolves under 20% of files.
- **Hidden provenance metadata** (C2PA) was recovered from inside every PNG, yielding per-image timestamps that identified records 1–13 as **pre-festival developer tests** (excluded from analysis; still processed).
- **100% of images processed, zero failures**; every analysed record belongs to exactly one emotional/non-emotional group (607/405) and one face/scene group (591/421) — the splits are groupings, not exclusions, and *all* groups are reported.
- 87 prompts repeat 2–42 times (336 records) — used both as a natural consistency experiment and for a deduplication robustness check.

2 Pipeline architecture

Following the supervisor's direction, the pipeline never asks a vision model for an emotion. Instead it *describes* then *extracts*:

Stage	Component	Output
0 · Split	GoEmotions (prompts) / BLIP-VQA cues (images)	emotional? · face?
1 · Describe	**BLIP-VQA**: expression, action, posture, scene	neutral description
2 · Extract	**Zero-shot NLI** (facts → feeling), both sides	top-3 emotions
3 · Compare	**Similarity ladder**, not exact match	4 metrics + null z

The similarity ladder scores each pair at four granularities — valence–arousal proximity, valence sign, Ekman family, and probability-weighted top-3 overlap — and every metric is referenced to a **random-pairing null** (300 shuffles): a pair only counts as "aligned" to the extent it beats chance.

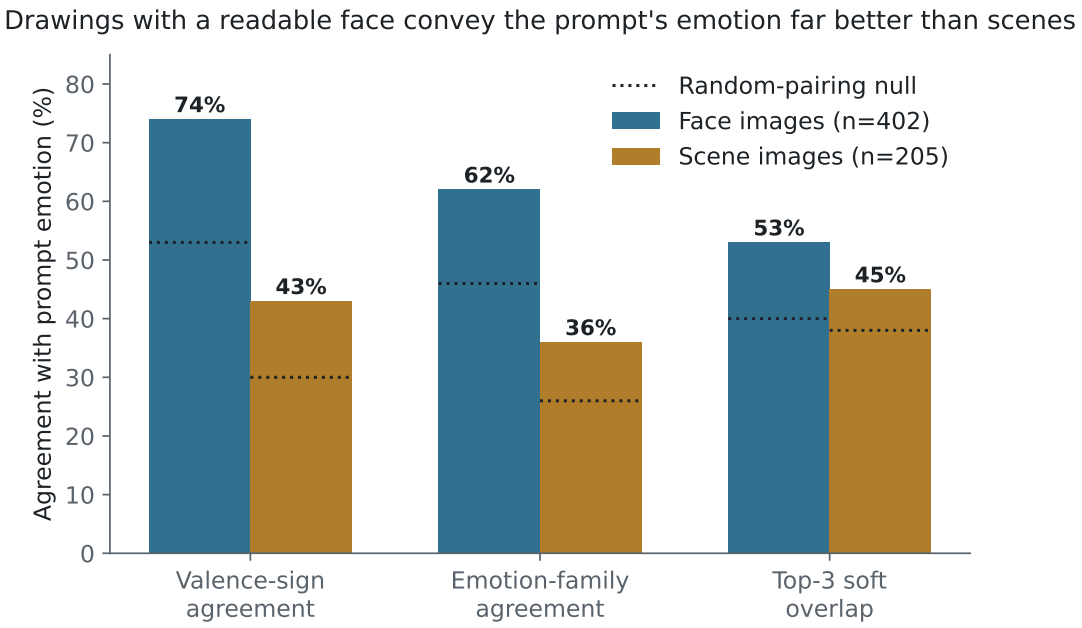
2.1 Engineering the pipeline: what was tried and rejected

Each link was tested *before* being trusted; the failures shaped the final design and are findings in their own right:

Candidate	Why rejected (evidence)
GoEmotions on image descriptions	Collapses to *neutral* 0.95+ — it needs first-person feeling words; scene descriptions have none.
Single BLIP caption → NLI	Captions too terse ("a man with hands in the air") — NLI collapses to *surprise*.
Qwen2.5-VL-3B describer	7 GB fp16 exceeded laptop RAM; thrashed into swap, zero progress in 4.5 h. Machine supports only sub-1B describers.
Florence-2 describer	Incompatible with transformers 5.14 (`forced_bos_token_id` failure in remote code).
BLIP-VQA (chosen)	∼0.4B, native, ∼1.5 s/record; its *"what is the facial expression?"* answer supplies exactly the affective cue NLI needs.

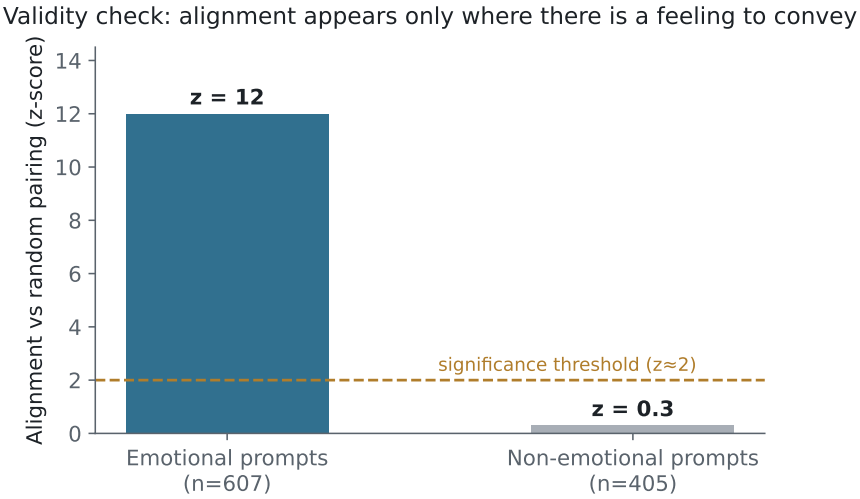
3 Findings

3.1 Finding 1 — faces carry the emotion; scenes carry far less



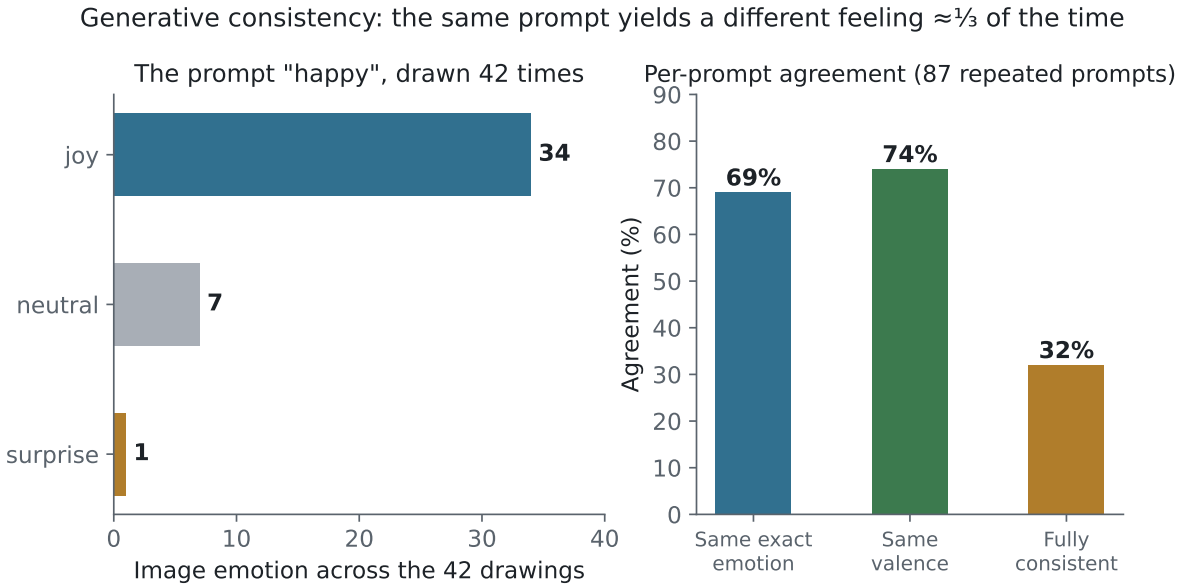
Face drawings (n=402) agree with the prompt's valence 74% ($z = +10.4$) vs 43% for scenes (n=205, $z = +5.8$); family agreement 62% vs 36%. The gap persists on deduplicated prompts (face $z = +9$). *Caveat (stated honestly):* the face/scene rule derives from the same description that feeds emotion extraction, so the split is partially entangled with the measure; the null-referenced gap remains large under every check.

3.2 Finding 2 — the validity check passes



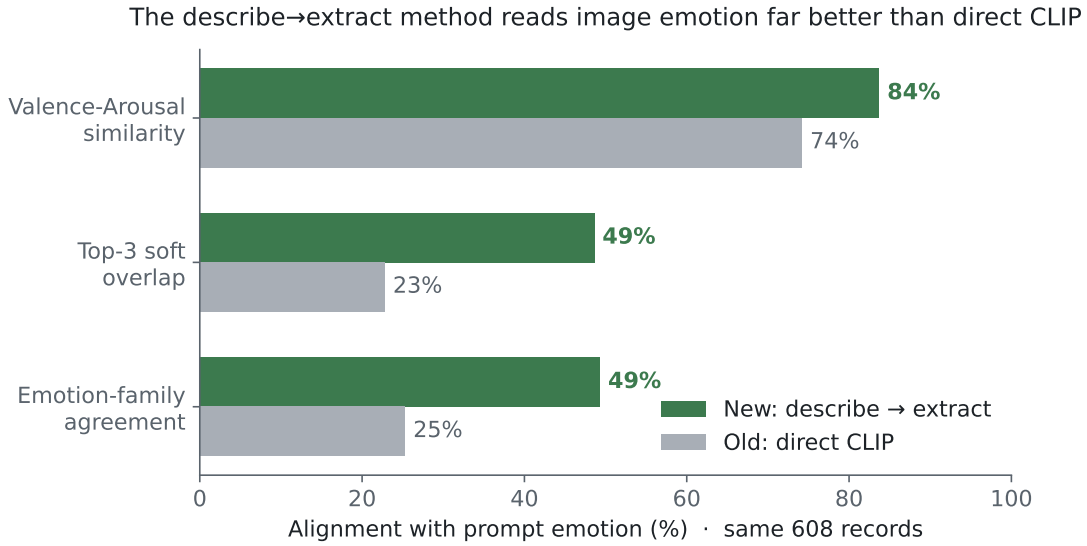
Emotional prompts align at $z \approx +12$; non-emotional prompts ("pigs", "mickey mouse") align at $z \approx +0.3$ — chance. The instrument detects emotion only where emotion exists, which is precisely how a trustworthy instrument should behave.

3.3 Finding 3 — generative emotion is unstable



Given an *identical* prompt, the model conveys the same emotion only 69% of the time (valence 74%); just 28 of 87 repeated prompts are fully consistent (these pool corpus-wide repeats — same text from different visitors and consecutive re-generations — the generator's input being identical either way). "happy" ×42 produced 34 joy, 7 neutral, 1 surprise.

3.4 Finding 4 — the describe-then-extract method wins



On the same 608 records, describe→extract roughly **doubles** both family agreement and top-3 overlap versus direct CLIP (the overlap figure corrected in the final audit — a label-space mismatch had deflated the old method's score) — empirical support for the supervisor's proposed architecture.

3.5 What is deliberately *not* claimed

- **"Positive emotions transfer better than negative"** — retracted after adversarial review: within a single-valence subset the sign/family null is degenerate ($z = 0$) and the raw gap is a base-rate artifact. The two automatic readers disagree in *opposite* directions (CLIP skews negative on line art; BLIP-VQA reads 66% of faces as smiling), so only the planned **human-rating study** can adjudicate.
- Fear, disgust and surprise are poorly captured by both readers; strong claims are restricted to the valence axis and the face/scene contrast.

4 Verification and robustness

- **Null baselines everywhere.** All metrics referenced to 300 random pairings; headline z range +4 to +16.
- **Adversarial statistical review.** An independent reviewer re-derived the analysis; its corrections (degenerate nulls, missing per-metric baselines, an overstated scene claim, undisclosed duplicates) were all applied.
- **Deduplication.** 249 duplicate-prompt records disclosed; every headline re-run on 763 distinct prompts — face finding robust.
- **Data integrity.** 1,025 unique ids, all image paths verified, all emotion fields well-formed, zero processing failures.
- **Automated test suite** (`test.sh`): environment, data, analyses, live pipeline, compilation — **18/18 PASS**.

5 Final audit

A six-dimension audit (code correctness; claim reproduction; document consistency; reproducibility; legacy-code health; deep data integrity) was run as the final step, and **every finding was fixed and re-verified**. The material findings and their resolutions:

Finding (severity)	Resolution
Label-space mismatch deflated the old method's top-3 overlap: 15% → true 23% (major)	Canonicalised `happiness→joy`; "tripled" corrected to "roughly doubled" in every document. The new-beats-old conclusion is unchanged.
Two scripts silently included the 13 pre-test records (major)	Pre-test filter added; affected table now $n = 984$ (values shift $\leq$0.4pp; no finding changes).
README and four dashboards presented the superseded direct-CLIP era as current (critical/major)	README rewritten around the final architecture; stale dashboards and write-ups moved to `outputs/archive/` with visible supersession banners.
Face rule missed multi-word answers ("man is smiling") — ∼2.5% leakage (minor)	Token-intersection matching; split refreshed (591/421), findings re-verified.
No pinned dependencies; setup script bypassed the venv; figures module missing from it (major)	`requirements.txt` (55 pinned packages); `setup.sh` now creates the venv and installs from it; figure generation added to the test suite.
Fragile old-baseline dependency (a legacy command would overwrite it) (major)	Baseline frozen as `results_clip_v2_baseline.json`; comparison script points at the frozen copy.
"Repeated-prompt runs" phrasing conflated corpus-wide repeats with consecutive runs (minor)	Reworded throughout; the pooling is now stated explicitly alongside the stricter consecutive-run count.
Assorted: two case-mismatched image paths (Linux-portability), unnormalised cached family scores, stale-cache crash path, misleading comments (minor)	All fixed; per-record paths verified against the on-disk listing; cache normalised and guarded.

The audit also *confirmed*: every published number reproduces exactly on a fresh run (deterministic, seeded); the null/z machinery is statistically sound; the data chain is airtight (1,025 unique ids, all paths resolve, zero malformed records); and the legacy baseline still runs. After fixes, the expanded automated test suite passes in full.

6 Limitations and next steps

- **Reader bias is the binding limitation.** BLIP-VQA's positivity bias and CLIP's negativity bias bound what automatic measures can claim; the **human-rating study** (ethics application drafted) is the designed adjudicator.
- Valence–arousal uses emotion-label centroids; a continuous lexicon (NRC-VAD) is a ready upgrade.
- The face/scene rule is description-derived; a dedicated detector would decouple it.
- Hardware bounded model choice (sub-1B describers); results are nonetheless statistically strong.

7 Repository guide

File	Role
describe_extract.py	The pipeline (crash-safe, resumable)
analysis_final.py	Final analysis: splits, similarity ladder, nulls, dedup
data_audit_consistency.py	Data accounting + repeated-prompt consistency
similarity_v2.py	Old-vs-new method comparison
make_figures.py	All report figures
test.sh	28-check automated test suite
emoviz/	Core package (loader; earlier baseline reader)
outputs/RESULTS_final.md	Verified findings write-up
outputs/describe_extract.jsonl	Full per-record pipeline output (1,025)
outputs/analysis_final.csv	Per-record analysis table (1,012)
PIPELINE_DESIGN.md	Architecture options & decisions

Prepared as technical documentation for the dissertation project. All numbers trace to re-runnable scripts in the repository; every figure regenerates from make_figures.py.